Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

A Hybrid AI-Enhanced Transaction Monitoring Framework for Fraud Detection

Archita (2210991335)
Sonali (2210992397)
Anshika (2210991303)

Supervised By
Dr. Isha Kansal

Chitkara University Institute of Engineering and Technology, Punjab, India

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Abstract

Background:

- Rapid growth of digital payments (Unified Payments Interface (UPI), Net Banking, credit/debit cards) has increased fraud risks worldwide.
- In 2025, over 425 million accounts were compromised -- 13 persons every second (Surfshark).
- Traditional rule-based systems generate high false positives and miss novel attacks.

Keywords: Fraud, AI (Artificial Intelligence), Transaction, Cybersecurity, Hybrid

Proposed Solution -- Hybrid AI Framework:

- Combines rule-based logic, machine learning, anomaly detection, and graph analysis.
- Designed for real-time detection in digital financial systems.
- Provides instant alerts with minimised false positives.

Key Result:

- Hybrid framework achieves 99.5% accuracy, outperforming all individual models.
- Compliant with RBI and Anti-Money Laundering (AML) regulatory frameworks.

Introduction

Problem Context:

- Indian digital payments (UPI, Net Banking) have grown exponentially but introduced new vulnerabilities.
- Attackers exploit: Phishing, SIM Swap, QR Hijacking, Social Engineering, and Mule Account networks.
- UPI's real-time transaction nature makes fraud recovery and detection highly challenging.

Limitations of Existing Approaches:

- Rule-Based Systems: Predefined thresholds lead to high false alerts and low accuracy.
- ML-only models: Improved detection but lack explainability required by regulators.
- Neither approach captures coordinated fraud across interconnected account networks.

Research Objective:

- Design a hybrid AI system combining rules, ML, anomaly detection, and graph analysis.
- Achieve high accuracy, low false positives, real-time detection, and RBI/AML compliance.

Literature Review

AI-Based Fraud Detection:
- Bhattacharyya et al. [1]: Random Forest outperforms LR and SVM on financial datasets.
- Dal Pozzolo et al. [2]: Ensemble methods with preprocessing address class imbalance.
- Whitrow et al. [5]: Transaction aggregation and feature engineering improve detection.

Deep Learning Approaches:
- Jurgovsky et al.: RNNs capture sequential transaction behavior.
- Fiore et al. [7]: GANs combined with ML outperform individual methods.
- Shohan et al. [8] (2024): Hybrid model with multi-class fraud classification.

Graph-Based Detection:
- Weber et al. [4]: Graph Convolutional Networks identify suspicious account links.
- Pandit et al. [3]: Graph mining detects fraudulent communities (mule networks).

Research Gaps Identified:
- No real-time UPI-scale deployment | High false positives | Explainability deficit
- Limited network-level analysis | Data imbalance | Regulatory compliance gap

Dataset & Preprocessing

Dataset Overview:

- Digital financial transaction records with binary classification: Normal (0) vs. Fraud (1).
- Features mirror real-life UPI, card, and wallet payment attributes with significant class imbalance (few fraudulent vs. many legitimate transactions).

Key Features: Transaction ID | Amount | Time | Device ID | IP Address | Location | Transaction Type | Beneficiary ID | Transaction Count

Preprocessing Pipeline:

1. Remove data-leakage identifiers
2. Handle missing values -- mean/median (numerical), mode (categorical)
3. Encode categoricals -- Label Encoding / One-Hot Encoding
4. Feature Scaling: x_scaled = (x - mean) / std_dev
5. Handle class imbalance -- SMOTE / cost-sensitive learning
6. Stratified Train-Test Split: 80% training / 20% testing

Hybrid AI Framework Architecture

Four Integrated Components:

1. **Rule-Based Engine**
   - Enforces RBI / AML compliance thresholds (limits, frequency, geographic anomalies)

2. **Machine Learning Layer**
   - Logistic Regression --> Random Forest --> XGBoost

3. **Anomaly / Divergence Detection**
   - Unsupervised statistical methods flag unusual patterns

4. **Graph-Based Analysis**
   - Nodes = Accounts, Edges = Transactions
   - Detects mule networks via degree centrality and clustering coefficients

Hybrid Risk Score = w1(Rule) + w2(ML) + w3(Anomaly) + w4(Graph)

Machine Learning Algorithms

Logistic Regression:

$P(y=1 \mid x) = 1 / (1 + e^{-(w^T * x + b)})$ Loss function: Cross-Entropy

Interpretable baseline; limited on complex nonlinear fraud patterns.

Random Forest:

$\hat{y} = \text{majority\_vote} \{ h1(x), h2(x), ..., hn(x) \}$

Ensemble of decision trees; handles high-dimensional data, reduces overfitting.

XGBoost:

$L(\phi) = \sum [l(y_i, \hat{y}_i)] + \sum [\Omega(f_k)]$

where $\Omega(f) = \gamma T + (1/2) \lambda \|w\|^2$

Gradient boosting with regularization; highest individual model accuracy.

Evaluation Metrics:

$\text{Accuracy} = (TP+TN)/(TP+TN+FP+FN)$ $\text{Precision} = TP/(TP+FP)$

$\text{Recall} = TP/(TP+FN)$ $\text{F1-Score} = 2*(\text{Precision}*\text{Recall})/(\text{Precision}+\text{Recall})$

Experimental Results

Experimental Setup:

Python 3.x | Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn

Pipeline: Data Load --> Preprocessing --> Model Training --> Evaluation --> Visualization

Model Performance Comparison:

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.94	0.93	0.92	0.92
Random Forest	0.98	0.97	0.97	0.97
XGBoost	0.99	0.99	0.98	0.98
Hybrid Framework	**0.995**	**0.99**	**0.99**	**0.99 [BEST]**

Ensemble methods significantly outperform Logistic Regression.

The Hybrid Framework achieves the best scores across all four metrics.

Results & Discussion

Key Findings:

1. Hybrid approach is optimal -- integrates rules, ML, anomaly, and graph detection.
2. False positives significantly reduced compared to conventional rule-based systems.
3. Graph analysis identifies mule account networks overlooked by traditional methods.
4. Effective preprocessing (SMOTE, stratified split) mitigates class imbalance.
5. Framework designed for real-time settings (UPI, Net Banking, Card Payments).
6. More transparent and RBI/AML-compliant than black-box ML-only models.

Feature Importance:

Top discriminators: Transaction Amount, Transaction Frequency,
Device ID, Geographic Location, Behavioral Patterns over time.

Limitations:

- Performance may vary across different real-world datasets.
- Binary classification only; limited real-time deployment testing.
- Higher computational cost than single-model approaches.

Conclusion

- Proposed a Hybrid AI Transaction Monitoring Framework for fraud detection in digital financial systems (UPI, Net Banking, Card Payments).

- Behavioral and transactional features (frequency, device, location) are critical discriminators between fraudulent and legitimate transactions.

- Effective preprocessing and class-imbalance handling (SMOTE) are essential for reliable model generalization on skewed datasets.

- Ensemble methods (Random Forest, XGBoost) far outperform Logistic Regression in capturing complex, nonlinear fraud patterns.

- The Hybrid Framework achieves 99.5% accuracy with minimal false positives, and is compliant with RBI and AML regulatory requirements.

Future Scope

1. Multi-class fraud classification
   Extend beyond binary detection to classify specific fraud sub-types (Phishing, SIM Swap, QR Hijacking, Mule Network, etc.).

2. Real-time banking system deployment
   Integrate with live UPI and Net Banking pipelines for production-grade detection.

3. Blockchain integration
   Combine with distributed ledger technology for immutable transaction audit trails, enhancing both security and transparency.

4. Lightweight / edge models
   Develop ultralight model variants for higher processing speed on low-resource devices.

5. Advanced deep learning frameworks
   Incorporate Transformers and Graph Neural Networks for further accuracy gains.

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Thank You